

RESEARCH ARTICLE

3D Printing Process and Local Preheating Technique Study Using a Laser-Based Method

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ABSTRACT

This study addresses the challenge of enhancing inter-filament bonding in Fused Filament Fabrication (FFF) 3D printing, where conventional methods often result in suboptimal mechanical strength or structural distortions. A novel local preheating technique was introduced using a laser-based approach to precisely control the temperature of previously deposited filaments. An experimental setup integrating a 3D printer with a 30W diode laser and an optical system was developed to evaluate key parameters, including laser power, PLA color effects on laser absorptivity, and the relationship between preheating temperature and bonding strength. Results demonstrate that preheating to 180°C increases mechanical bonding strength by up to 35% without distortion, highlighting the technique's potential to improve FFF part quality.

1 | Introduction

The Additive Manufacturing (AM) process comprises various techniques allowing the construction of three-dimensional parts of complex geometry that are difficult or impossible to manufacture using traditional manufacturing processes [1, 2]. Fused Filament Fabrication (FFF) is one of those AM processes used to produce prototypes in different industrial sectors such as aerospace, medical, automotive, etc. [3, 4]. In FFF, the most important materials for parts manufacturing are thermoplastic polymers [5]. Accordingly, several parameters affect the printing part quality [6, 7], like the temperature profile of the polymer filaments and consequently the inter-filament bonding [8–10]. It is therefore important to understand how the process parameters affect the evolution of the filaments temperature as mentioned [11–14].

The problems of effective bonding and reduced mechanical performance are a major concern of 3D-printed structures. When 3D-printing a given part, as the deposition progresses,

the melted filament is deposited onto filament that was previously deposited which is in the process of cooling. The contact between the melted filament and the previously deposited filament causes reheating of the latter. At the interface of adjacent filaments, the temperature rises above the crystallization temperature and proper bonding takes place. Therefore, to foster material crystallization at the interface between filaments and allow better molecular chain rearrangement during the deposition of the melted filament, the previously deposited filament temperature should be sufficiently high, probably about the crystallization temperature. A higher temperature of the previously deposited filament could cause structural distortions, while a lower temperature does not allow a proper molecular chain rearrangement, resulting in lower bonding of adjacent filaments [15].

Nowadays, one of the challenges in improving the bonding of 3D-printed parts concerns optimizing the temperature at the interface between the melted filament and the previously deposited filament. Therefore, controlling the cooling speed or temperature

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Summary

- Novel laser setup improves FFF 3D printing quality.
- Preheating to 180°C enhances bonding without distortion.
- PLA color affects laser absorption; black absorbs most.
- Laser preheating boosts 3D printing bond strength by 35%.

profile of the previously deposited filament is essential for interface bonding and the strength of the 3D-printed part [8]. A high cooling rate leads, on the one hand, to a decrease in the degree of crystallization, resulting in reduced polymer strength and rigidity, especially in the case of PLA; and, on the other hand, to poor interfacial bonding. Conversely, a low cooling rate causes instability in printed parts. Current FFF methods struggle with poor inter-filament bonding due to rapid cooling, which limits crystallization and mechanical strength. Global heating techniques, such as chamber heating, often result in warping or dimensional inaccuracies due to uncontrolled temperature distribution. Conversely, a local preheating technique addresses these limitations by selectively raising the temperature of previously deposited filaments, fostering molecular diffusion and bonding at the interface without compromising the overall structure.

This study focuses on the thermal phenomena involved in the FFF 3D printing process. These phenomena play a key role for inter-filament bonding and dimensions of 3D-printed parts. To improve the adhesion between the filaments and strengthen their interface, an original laser-based preheating technique is used to locally preheat the already deposited filament, just before its contact with the molten filament. This precise technique, based on temperature control, ensures effective welding of the filaments at a temperature higher than the polymer's crystallization temperature.

2 | Local Preheating Technique—State of the Art

In the previous work of authors [16], the effect of previously deposited filament temperature on interface bonding improvement was studied. Several printing tests, aiming to maintain the temperature of the previously deposited filament higher than the crystallization temperature, were carried out. To meet such a condition, it was necessary to minimize the cooling time (Δt). This time represents the difference between the melted filament deposition and the previously deposited filament times. Given that the cooling time (Δt) is inversely proportional to the printing speed, the latter was increased till a given limit depending on the printing quality. The results indicated a 23% improvement in inter-filament bonding for small temperature differences (ΔT) between the previously deposited and the melted filaments.

Several numerical and experimental studies have investigated the thermal behavior of filaments in FFF, emphasizing the critical role of thermal history in determining adhesion properties. Costa et al. [17] developed a heat transfer model to predict temperature profiles and adhesion quality during filament deposition, showing good agreement between numerical and

experimental results. Sun et al. [15] explored bond formation mechanisms in ABS filaments, demonstrating that temperature variations under convective conditions significantly affect heat transfer and bond strength. Zhou et al. [18] examined how nozzle and platform temperatures, extrusion speed, and layer thickness influence thermal profiles and bonding. However, these studies highlight a key challenge: the thermal history of previously deposited filaments—dictated by cooling rates and reheating cycles—directly impacts molecular diffusion and entanglement across interfaces, affecting not only bond strength but also interlayer fracture toughness. For instance, rapid cooling reduces crystallization and limits chain mobility, leading to weaker interfaces prone to delamination, as evidenced by lower fracture toughness values reported in PLA parts. Conversely, excessive reheating can degrade polymer chains or cause dimensional instability, further complicating adhesion optimization. Recent research also shows that cyclic cooling and reheating influence sintering through temperature-dependent viscosity changes, yet few studies address how these thermal dynamics affect inter-layer fracture toughness—a critical metric for part durability under stress. They noted that insufficient thermal energy at the interface reduces toughness by limiting interdiffusion, underscoring the need for precise temperature control. Thus, current methods struggle to balance thermal history to enhance both adhesion and mechanical integrity, motivating the focus on local preheating to target these interfacial properties effectively. Besides this, inter-layer adhesion has been widely investigated.

Recent research demonstrated that cyclic cooling and re-heating of filaments during FFF has an impact on sintering by considering the effect of temperature on viscosity [19, 20]. Besides this, inter-layer adhesion has been widely investigated. Yin et al. [21] investigated, both numerically and experimentally, the effect of process parameters on the bonding strength of multi-materials, and they found that the variation of temperature profile has an impact on the mechanical behavior of the printed parts. Consequently, they focused on the improvement of inter-layer bonding by the addition of bimodal blends to the polymeric parts and found that Low Molecular Weight (LMW) additives reinforce interfaces and reduce anisotropy [22, 23]. These efforts led to the application of novel materials, using acrylonitrile-butadiene-styrene (ABS) as a matrix, thermoplastic polyurethane (TPU) as an additive, allowing 3D printing without platform heating [24].

To improve bonding in the 3D-printed part, other researchers focused on preheating the previously deposited filament to a given temperature before depositing the melted filament. Among these, Partain [25] used a nozzle to blow hot air onto the fabricated part to decrease the cooling rate of the previously deposited filament. While hot air preheating slows cooling, its lack of spatial precision can overheat adjacent areas, risking distortion and limiting its effectiveness for complex geometries. Kishore et al. [26] exerted an infrared ray to evaluate the strength of fabricated parts. Infrared heating enhances strength but often causes surface damage due to uncontrolled energy distribution, reducing its practicality for high-quality parts. Despite obtaining interesting results, a surface damage was observed because of the infrared ray radiation. A local preheating laser was also applied to the layered zone near the extruder outlet to increase the interface temperature between filaments. The preheated sample had more strength, elasticity, and toughness, but the material flow

affected the quality of printed parts [27]. Shih et al. developed a distinct method during their work [28]. According to their findings, conventional approaches involving heating the printed material tend to introduce defects in the form of distortion or warping, compromising the expected precision. They highlighted the time-consuming nature and complexity associated with most heating methods. In response, they propose the Cold Plasma Treatment (CPT) solution, addressing these challenges. This technique employs a high-voltage and high-current plasma discharge to modify the physicochemical properties of the part up to a depth of 50 nm. To achieve this, electrons are heated to 10,000°C and transported by a gas (helium, at room temperature) under atmospheric pressure. Their system comprises a stainless-steel electrode positioned 1 mm from the part, secured by an adjustable platform, connected to a nanosecond pulsar as a power supply, capable of reaching 30 kV, with a pulse frequency of up to 10 kHz and a pulse width of 3 ns. This process induces surface oxidation, enhancing wettability within the part and the polarity of the molecular structure, consequently improving diffusion and enabling the creation of new bonds. Stress tests were conducted considering two durations of plasma discharge: 30 s and 300 s. The results demonstrate a significant increase in tensile strength—by 134% for a discharge time of 30 s and by 63% for 300 s—compared to a sample untreated with CPT. This method proves to be very effective, showcasing current achievable performance. However, CPT significantly boosts tensile strength (up to 134% for 30 s exposure), but its reliance on high-voltage equipment and complex setup increases costs and reduces accessibility compared to the proposed laser-based approach, which achieves a 35% strength increase with a simpler, scalable system. Its primary drawback lies in the setup and device costs, which are potentially more intricate and less straightforward than the previous methods.

Moreover, the study undertaken by Rajpurohit et al. [29] serves as a valuable complement to the aforementioned investigation. They conducted a series of tests, systematically varying several printing parameters to assess their respective impacts on the bond strength. Their specific attention was directed towards

the deposited PLA layer height and the screen angle. Through a meticulous analysis of the test samples, they discerned that layer height exerts the most substantial influence on the tensile strength, followed by the screen angle. Furthermore, they pinpointed that the optimal combination of layer height and screen angle occurs at 100 µm and 0°, resulting in an approximately 10% increase in tensile strength.

Unlike these methods, the proposed laser-based preheating technique provides precise, localized temperature control, significantly improving the bonding strength without structural defects, offering a practical advancement for FFF applications.

3 | Local Preheating Technique—Experimental Setup

In this study, a laser beam has been used to locally preheat the previously deposited filament, shortly before depositing the melted filament. Thanks to this local preheating technique, it was sought to set and control the temperature at the interface between the previously deposited and melted filaments, thus (i) fostering a deep molecular rearrangement at the filaments interface and (ii) enhancing a high filament-to-filament bonding strength.

Following the above-mentioned explanations, an experimental setup comprising a 3D printer equipped with a 30 W diode laser emitting at 830 nm, coupled with 0.6 mm core diameter optical fiber, was built. To help focus the laser beam to a fine spot diameter and induce greater heat on the targeted surface, an optical system was designed using two collimating and focusing lenses. This optical system was positioned and attached to the nozzle NC axis thanks to a 3D printed customized support and directed towards zone A, slightly before the melted filament deposition zone to preheat the previously deposited filament to a desired temperature (see Figure 1).

To run properly this experimental setup, several actions were taken like: (i) calculating the necessary laser power allowing to

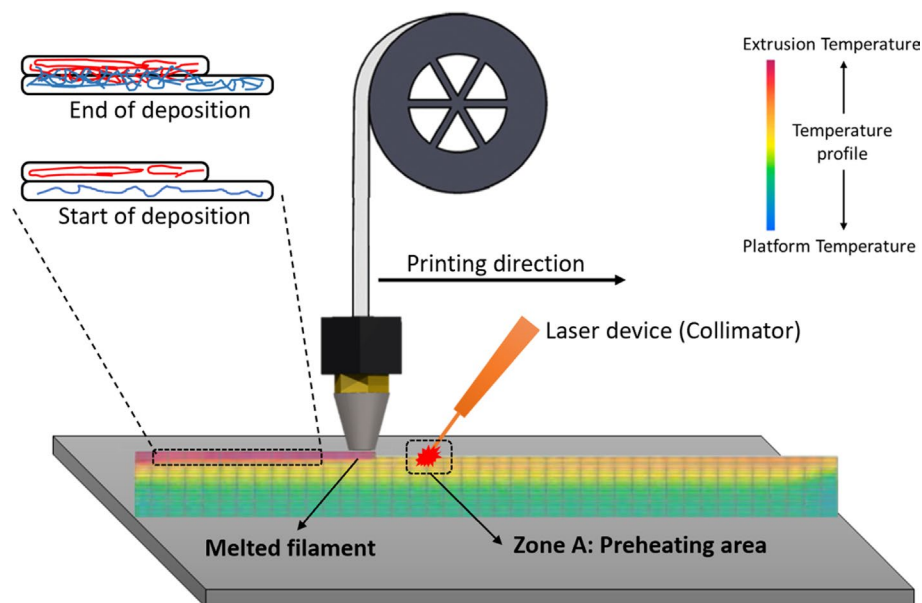


FIGURE 1 | Representation of the proposed solution.

preheat the PLA polymer to a given temperature, (ii) studying the effect of the PLA polymer color on the level of laser absorption, (iii) developing an automated solution for the laser on/off control so as to precisely control the duration of the laser shots, (iv) using a thermal camera to measure the infrared energy involved in PLA polymer preheating and then evaluating the preheating temperature. In what follows, the experimental setup is presented.

3.1 | Optical System Design

The applied optical system is designed to incorporate a tandem arrangement of two lenses within an optical tube. The initial lens, characterized by a focal distance denoted as f_1 and referred to as the collimator, serves the crucial role of receiving divergent rays emanating from the fiber. This collimator lens effectively transforms the incoming divergent rays into a beam characterized by parallel rays with infinite focal lengths. Consequently, this process mitigates the initial divergence of the beam, resulting in a more consolidated and directed optical output. The second lens in the optical system is a converging lens with a focal length denoted as f_2 . The primary function of this converging lens is to focus the collimated beam in a specific direction, thereby achieving a concentrated beam at a defined spot with a diameter d_{spot} . The optical system is designed to incorporate a tandem arrangement of two lenses within an optical tube. The initial lens, denoted as f_1 , serves the crucial role of receiving divergent rays emanating from the fiber. This collimator lens effectively transforms the incoming divergent rays into a beam characterized by parallel rays with infinite focal lengths, mitigating the initial divergence of the beam and resulting in a more consolidated and directed optical output. The second lens is a converging lens with a focal length denoted as f_2 . Its primary function is to focus the collimated beam in a specific direction, achieving a concentrated beam at a defined spot with a diameter d_{spot} . Through the careful orchestration of these optical elements, the system ensures an image of the output of the optical fiber on the surface of the material to be treated, with a magnification given by the ratio f_2/f_1 , as illustrated in Figure 2.

So, knowing that the diameter of the fiber is $d_{\text{fiber}} = 0.6$ mm, the magnification corresponding to the lens system would be calculated based on Equation (1):

$$\frac{d_{\text{spot}}}{d_{\text{fibre}}} = \frac{f_2}{f_1} \quad (1)$$

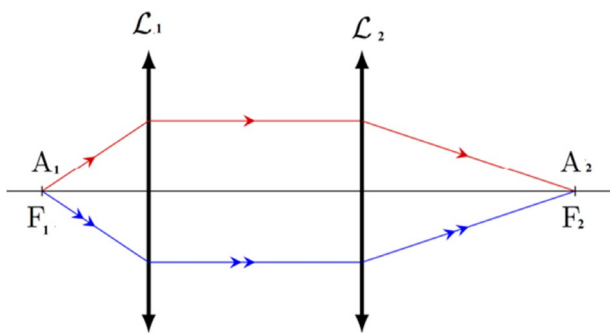


FIGURE 2 | Simplified schematic of the optical system.

The ThorLabs catalog provides us with a range of f_1 focal lengths for various types of lenses. So, to size the two lenses, the focal length f_2 corresponding to each value of f_1 in the range of available values has been calculated. Then, two lenses were chosen with a minimum couple ($f_1 = 35$ mm and $f_2 = 100$ mm) in order to ensure satisfactory compactness so as not to clutter the device. The spot diameter is then recalculated using the magnification law, leading to $d_{\text{spot}} = 1.71$ mm. Following this, a lens of 1-in. diameter with an Anti-Reflective Coating (ARC) guaranteeing a wavelength range that varies between 650 and 1050 nm was chosen, in compliance with the used laser (830 nm). Finally, a lens holder with an interior diameter $D_{\text{int}} = 30.5$ mm was selected. The detailed elements of the optical system are shown in Figure 3. This optical system is placed in a holding part designed to be fixed to the nozzle NC axis. This holding part allows a correct positioning of the optical system, directing the laser beam towards the previously deposited filament, slightly before the melted filament deposition zone. It follows from this design that the optical system and the nozzle move together, thanks to the NC nozzle axis, at a speed equal to the printing speed. This configuration with one laser spot positioned in front of the wire allows only simple shape parts printing, like the test sample used for this study.

3.2 | Numerical Simulation of the Required Laser Power

The objective was to determine the necessary laser power to achieve a maximum preheating temperature of 180°C. To that end, a numerical simulation was carried out using COMSOL Multiphysics (version 6.2, COMSOL AB, Stockholm, Sweden; www.comsol.com). This simulation was a temporal study of the heat transfer in a solid type, governed by the heat equation as expressed by Equation (2):

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p u \nabla T = \nabla \cdot (k \nabla T) + Q \quad (2)$$

For the simulation of the laser radiation, a solid block with dimensions $40 \times 10 \times 0.2$ mm has been modeled (Figure 4a), and the following boundary conditions have been defined:

- Heat transfer in the solid is governed by the heat equation (volume)
- Imposed 50°C temperature on the lower surface which is in contact with the platform (surface)
- Conductive heat losses on the side faces through the surrounding air (surface)
- Laser and melting filament heat inputs and heat losses on the upper face (surface)

Accordingly, the heat fluxes have been modeled as follows:

- Heat losses (W/m^2)

$$H.L = hC_v(T - T_{\infty}) + \epsilon\sigma(T^4 - T_{\infty}^4) \quad (3)$$

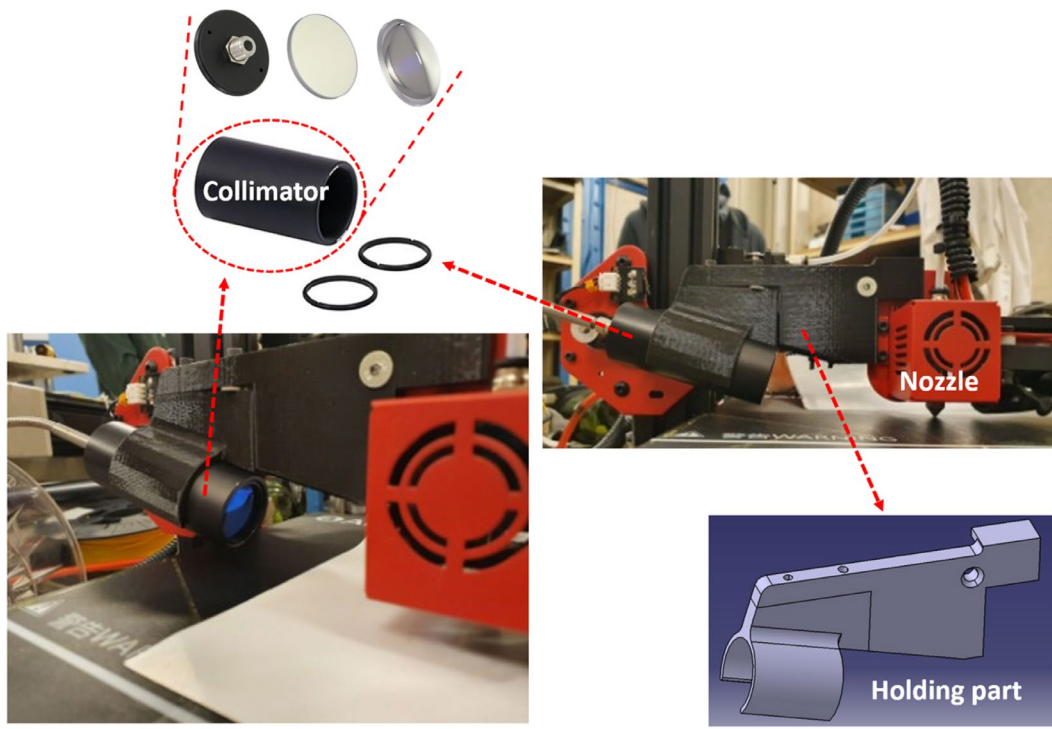


FIGURE 3 | Detailed representation of the laser local preheating set-up.

- Laser power (W/m^2)

$$L.P = \alpha \frac{P_{\text{laser}}}{\pi R_{\text{laser}}^2} \left((x - x_0)^2 + (y - y_0)^2 \leq R_{\text{laser}}^2 \right) \quad (4)$$

- Heat of filament (W/m^2)

$$H.F = k (T - T_{\text{filament}}) (-0.0005 \leq x \leq 0.0005) (y \leq y_0) \text{ and } y_0 = v * t \quad (5)$$

To proceed with the simulation, required data have been implemented as presented in Table 1.

For the purpose of this numerical simulation, a finer mesh on the upper part (Figure 4b) of the solid block was carried out, given that the phenomena of interest take place in this area. Finally, the simulation time was set to 2s, with a step every 0.05s, thus producing 40 images. The thermal results coming out of this simulation showed that for 1W laser power (Figure 4c,d), the maximum temperature reached at the upper part of the block approaches 400 K (127°C), which is equivalent to the crystallization temperature of the PLA. These numerical simulation results are only a rough approximation of the real laser power needed because the numerical model does not consider the laser absorption capacity of the material, which depends on the material's nature and color as well as the laser wavelength. Therefore, for a better estimation of the necessary laser power allowing an optimal preheating of the PLA material, experimental tests considering the absorption capacity for different PLA colors were carried out.

3.3 | Investigation on the Effect of PLA Color on Laser Absorption

Knowing about the “PLA laser absorptivity versus PLA color” allows to quantify both the absorbed, transmitted and reflected portions of the laser energy. Thanks to this characteristic, it is possible to calibrate precisely the laser power in order to reach the desired local preheating temperature. To study the laser absorptivity of the PLA material, a setup consisting of a laser beam, two reflective and diffusing spheres -Hulbrich spheres-, each instrumented with a photodiode giving a signal proportional to the reflected light inside the sphere, has been implemented as presented in Figure 5. The laser entering the two spheres strikes the photodiode at an angle such that the beam reflected by the sphere is completely diffused, and the light intensity within the sphere exhibits a homogeneous behavior. To determine the percentages of the reflected part of the beam and the transmitted one, the ratio between the photodiode voltages with and without samples was established.

First, an empty reference measurement was performed by directing the laser onto the lower sphere, creating a reference for the transmitted signal (S1). Subsequently, by positioning a 100% reflective mirror to redirect the entire signal into the upper sphere, a reference for the reflected signal (S0) was recognized. To determine the transmission, reflection, and absorbance of the material, a disk-shaped sample was placed in the setup and simultaneously the voltage in the upper sphere (S0') and the lower sphere (S1') was measured. Then, the transmission (T), reflection (R), and absorbance (A) of the material were derived by using the following equation:

$$R = \frac{S0'}{S0} \quad (6)$$

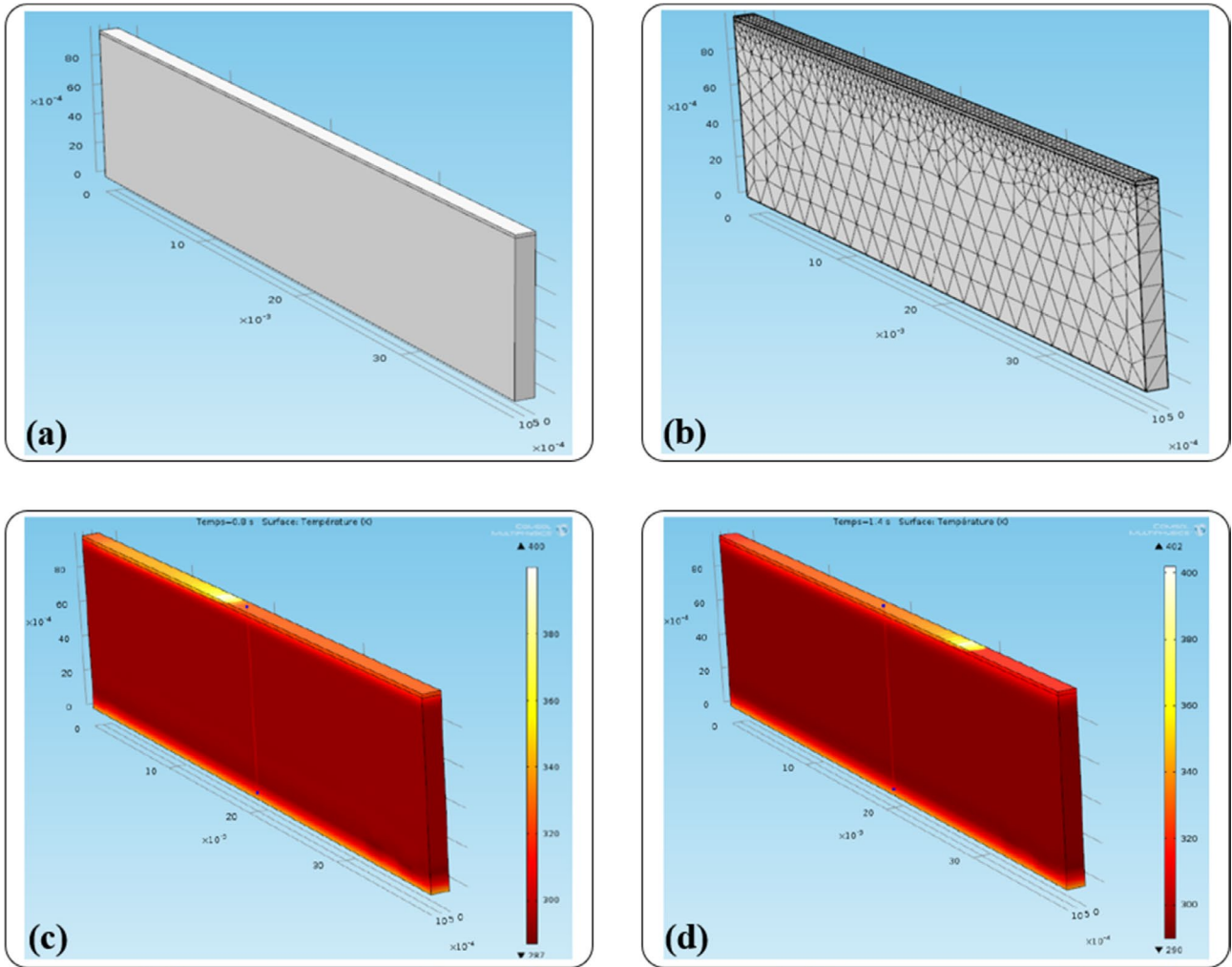


FIGURE 4 | Representation of the different steps of the numerical simulation: (a) 3D model of the solid block, (b) meshing of the 3D model, (c) temperature of the solid after 0.8s, and (d) temperature of the solid after 1.4s.

TABLE 1 | Data used for laser power calculation.

Parameters	Value
T_{∞} (K)	293.15
T_{filament} (K)	483.15
ε (emissivity)	0.92
σ ($\text{W/m}^2\text{K}^4$)	5.67×10^{-8}
R_{laser} (Laser radiation radius—mm)	0.85
Laser position (x_0 —mm)	0.5
Velocity of laser displacement (v —mm/s—1)	20
H ($\text{W/m}^2\text{K}$)	15
α	0.3
K (thermal conductivity—W/m K)	0.13
ρ (thermal conductivity—Kg/m ³)	1250
C_p (Heat capacity at constant pressure—J/Kg K)	1800

$$T = \frac{S1'}{S1} \quad (7)$$

$$A = 1 - R - T \quad (8)$$

For the purpose of this PLA absorptivity study, several PLA disk-shaped samples of different colors (White, Yellow, Red and Black) and thicknesses (1, 3, and 5 mm) were printed. The diode laser power was set around 1 W.

The test results allowed us to get the PLA absorptivity and transmittance as a function of thickness and color. The color of the PLA material significantly influences its laser absorption. Darker colors, such as black or navy blue, exhibit higher absorptivity rates because they absorb a broader spectrum of light wavelengths compared to lighter colors. The increased absorption leads to higher material temperatures. Consequently, darker PLA materials experience more pronounced thermal effects. Conversely, lighter-colored PLA materials, such as white or pastel shades, exhibit lower absorptivity. These colors reflect a larger proportion of light, thereby absorbing less heat. The impact of filament color and the thickness of 3D-printed parts on absorptivity is illustrated in Figure 6. Notably, black color exhibits a 75% absorption level and a consistent behavior across various thicknesses, while lighter colors show slight changes in absorption percentage, ranging between 10% and 30%, based on material thickness and color.

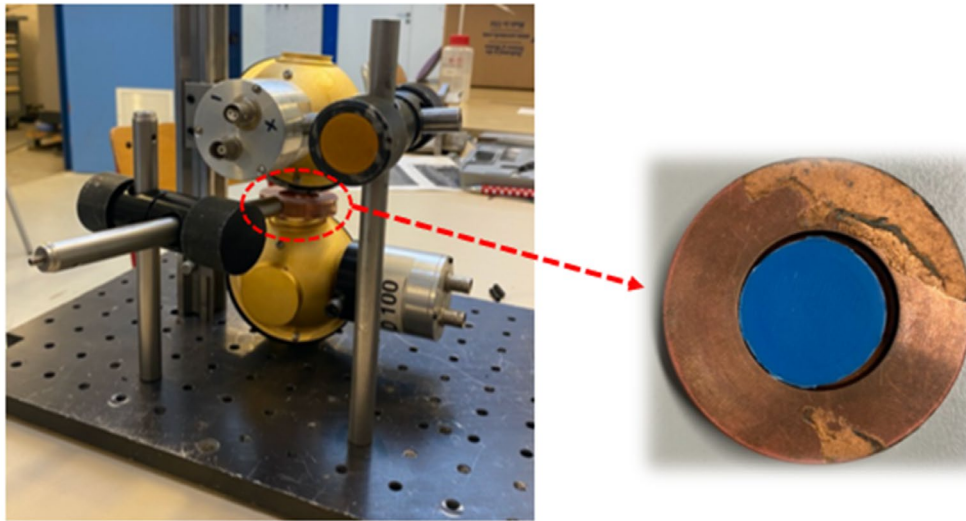
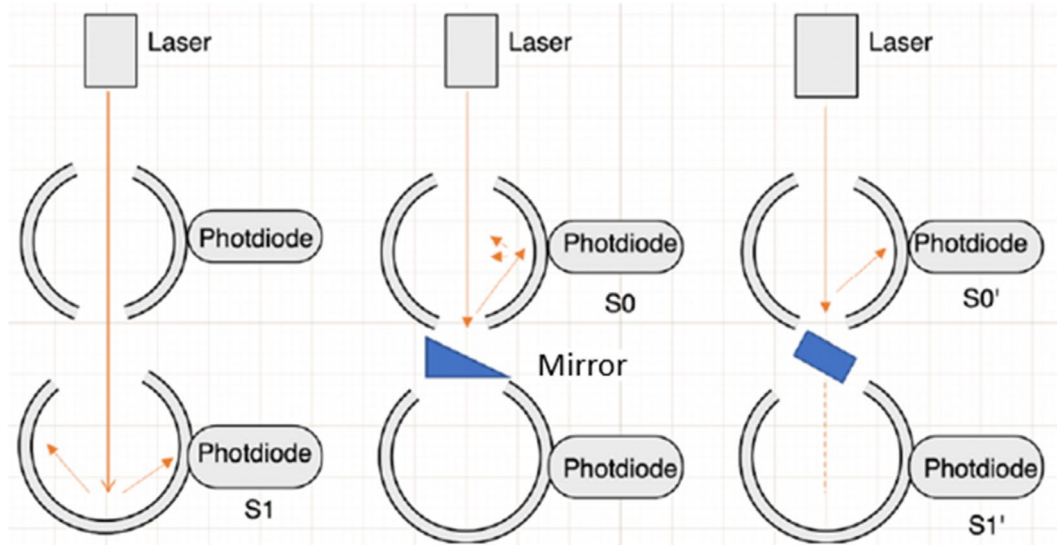


FIGURE 5 | Detailed representation of the proposed set-up for the study of polymer color effect on laser absorption.

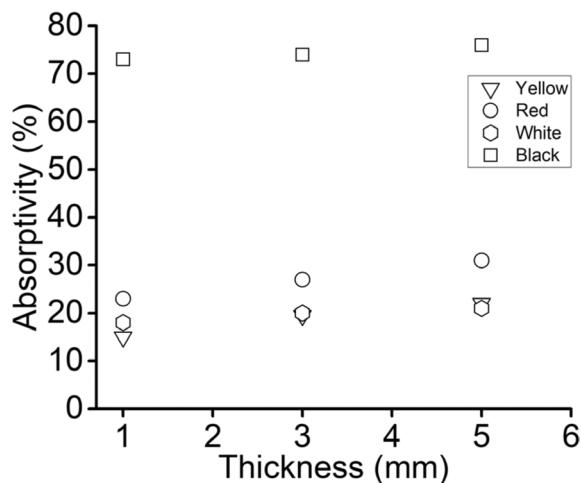


FIGURE 6 | PLA color and laser absorptivity.

3.4 | Automation of Laser On/Off Control

When printing the solid block using the preheating technique, it is essential to activate the laser beam intermittently. Since the printer deposits the melting filament in only one direction, the laser beam should not remain active during the “return” phase, as this imparts excessive energy to the polymer leading to measurement bias. Therefore, the laser beam should only be activated to preheat the material before the melting filament is deposited. To ensure regular activation, an automated system for laser ignition was designed and built (Figure 7) acting as follows: (i) A Hall effect sensor detects the nozzle return to its starting position; (ii) subsequently, an Arduino circuit sends the laser ignition command via its serial port; (iii) after a latency period, the extinction command is issued. This sequence is repeated until the whole block printing is accomplished. The said automated system allows the adjustment of the electrical intensity supplied to the laser with increments of 0.5A.

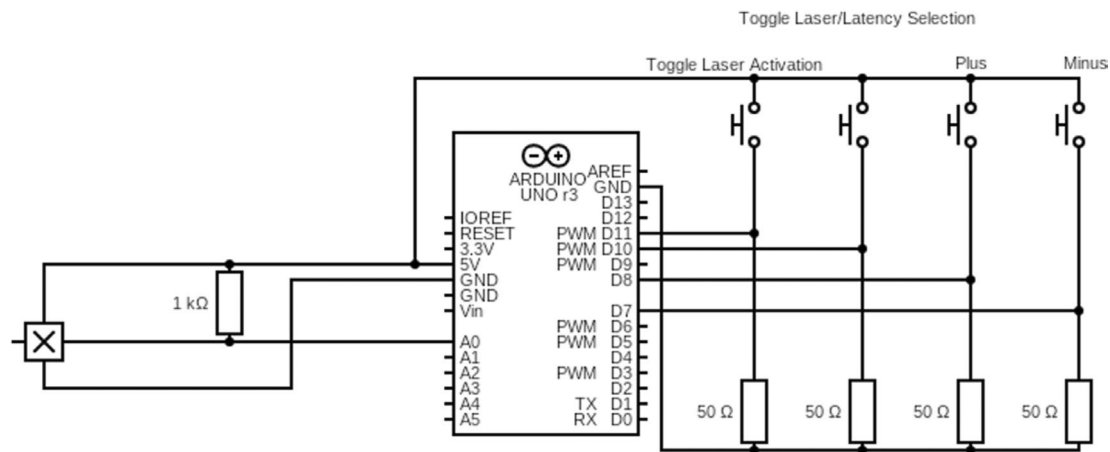


FIGURE 7 | The Arduino circuit and hall effect sensor.

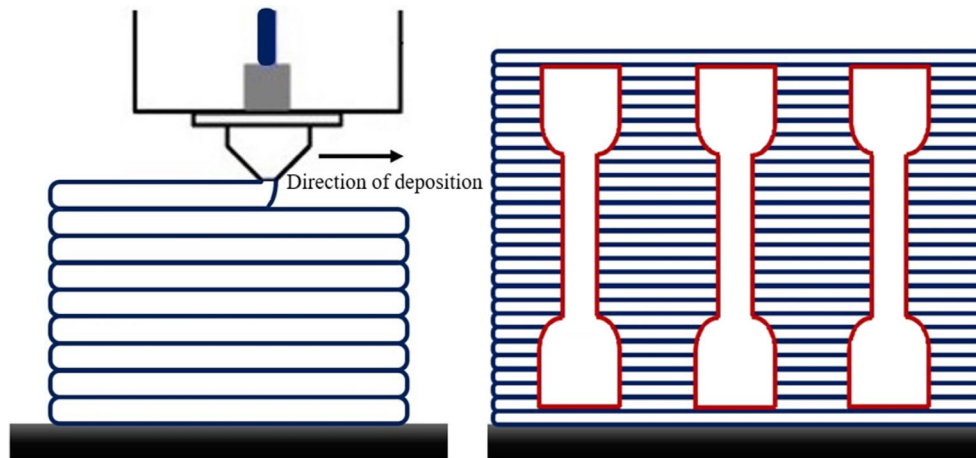


FIGURE 8 | Solid block filaments deposition procedure and sample shape.

4 | Materials and Characterization Techniques

4.1 | Materials, Test Design, and Sample Printing

The material used in this study for manufacturing the solid blocks was Polymaker PolyLite PLA filament (Polymaker, Shanghai, China) with a diameter of 1.75 ± 0.01 mm and a density of $\rho = 1.24 \text{ g cm}^{-3}$. It is a semi-crystalline polymer with a melting point of 150°C . The filament was stored in a sealed container with silica gel desiccant, maintaining a relative humidity of approximately 30% to prevent moisture absorption prior to printing, and was supplied in a black color. Several samples were printed to assess the efficiency of the laser preheating technique. The model corresponding to the solid block illustrated in Figure 8 was designed using the FreeCad software and then exported in STL format to be loaded into the Flash Print software, which generates the printing path [30].

The printed blocks were manufactured using the designed experimental setup, composed of (i) a desktop 3D printer fitted with a single nozzle ($d = 0.4 \text{ mm}$) printing head and a temperature-controlled atmosphere, and (ii) a laser fiber with its optic system. The test case was built considering the values for the processing variables (Table 2) that are commonly used in the desktop 3D printer to ensure a good quality part in terms of filaments bonding and mechanical strength [8, 31–34]. Regarding

TABLE 2 | Applicable process parameters for 3D printing of samples.

Parameter	Value
PLA liquefier temperature ($^\circ\text{C}$)	210
Platform temperature ($^\circ\text{C}$)	50
Printing speed (mm s^{-1})	20 and 40
Laser intensity (A)	15 to 35
Layer height (mm)	0.2
Filament cross-section	Circular

the local preheating technique, the laser intensity was increased from 15A to 35A, with 0.5A increments. In order to monitor the temperature of the block during all the local preheating/printing process, an infrared thermal camera was used to record the thermal state/surface temperature of the block and its filaments.

4.2 | Mechanical Characterization: Quasi-Static Tensile Tests

Tensile tests have been performed on the INSTRON 4301 machine. The specimen geometry used for quasi-static tensile tests is illustrated in Figure 8. However, based on the dimensional

change that occurs during the process and specimen preparation, a digital caliper was implemented to precisely measure the required dimension. All calculations regarding mechanical behavior have been proposed as mentioned. The loading velocity was equal to 1 mm/min^{-1} .

4.3 | Thermal Measurement Calibration and Error Analysis

Temperature measurements were conducted using an infrared thermal camera to monitor the local preheating temperature of the PLA solid blocks during printing. To ensure measurement accuracy, the camera was calibrated using a blackbody reference source with a temperature range of 50°C to 300°C , encompassing the experimental conditions (50°C to 270°C). The calibration procedure involved positioning the camera at a fixed distance from the blackbody, setting the emissivity to 0.92, and recording temperature readings at 10°C intervals. The camera's readings were compared to the blackbody's known temperatures, and a calibration curve was established to correct for systematic deviations, achieving a linearity within $\pm 1^\circ\text{C}$ across the range. Error analysis was performed to quantify measurement uncertainties. Key sources of error included variations in PLA emissivity (± 0.02 due to surface finish and color consistency), thermal camera resolution ($\pm 2^\circ\text{C}$ as per typical manufacturer specifications), and ambient temperature fluctuations ($\pm 1^\circ\text{C}$ during experiments).

4.4 | Thermal Characterization: Differential Scanning Calorimetry Tests

Differential Scanning Calorimetry (DSC) was accomplished using a TA instruments Q1000 (New Castle, USA). Extracted samples ($\sim 5.5 \text{ mg}$) sealed in aluminum pans and heated from ambient temperature to 210°C with heating rates ranging from 5°C min^{-1} to 2°C min^{-1} for determining crystallization and the related temperature of the filaments.

5 | Results and Discussion

5.1 | Preliminary Observations

5.1.1 | Influence of PLA Color on Laser Absorption

To master local laser preheating, the first actions consisted of building a specific experimental 3D printing device. The necessary laser power, to allow the PLA material to reach a maximum preheating temperature of 180°C , is first evaluated, numerically and experimentally. The objective was to study the bonding improvement versus preheating temperatures reaching crystallization temperatures and possibly beyond that, till 180°C . Knowing about the effect of the material color on laser absorption, efforts have been made to investigate the PLA laser absorptivity. The obtained results showed that the PLA color significantly influences laser absorption, as illustrated in Figure 6. Darker colors, such as black or navy blue, exhibit higher absorptivity rates due to a broader spectrum of light wavelengths absorption, leading to higher PLA material temperatures. Consequently, darker PLA materials experience more pronounced thermal effects. Conversely, lighter-colored PLA

materials, such as white or pastel shades, exhibit lower absorptivity. These lighter colors reflect a larger proportion of light, thereby absorbing less heat. The results in Figure 6 also show that black color has an absorption level of approximately 75% and consistent behavior across various thicknesses, while lighter colors have slightly lower absorption percentages, ranging between 10% to 30%. Following these results, black PLA was chosen for the local preheating technique study.

5.1.2 | Crystallization

The PLA used in this study is a semi-crystalline polymer with a slow crystallization rate. In this case, if the cooling rate is high, the polymer cannot fully crystallize during cooling. As a result, it does not properly follow the cooling process and remains either completely amorphous or only partially crystallized. This phenomenon can significantly impact the mechanical properties of the polymer.

To study the influence of cooling rate on crystallization, the DSC test was conducted with a heating rate of 5°C min^{-1} from room temperature to 210°C . The sample was then cooled at the same rate back to room temperature. The result is shown in Figure 9a. On the DSC diagram, during heating, a glass transition temperature (T_g) is observed at around 58°C , followed by a melting peak at approximately 149°C . During cooling, however, there is almost no crystallization peak. A T_g is observed in the same range as during heating.

To determine the critical cooling rate at which the polymer can crystallize, several tests were performed at various cooling rates. The results are shown in Figure 9b. A crystallization peak appears when the cooling rate is reduced to 4°C min^{-1} or lower. This peak becomes more pronounced as the cooling rate decreases. This experiment provides crucial information. Since the cooling rate during part manufacturing by FFF is much higher than 5°C min^{-1} , the molten filament does not crystallize upon cooling. Instead, it remains completely amorphous.

5.2 | Laser Intensity and Local Preheating Temperature

In order to study the relationship between laser intensity and local preheating temperature, several solid blocks were printed using laser intensities ranging from 15A to 35A. The local preheating temperatures were measured using an infrared thermal camera, with an uncertainty of $\pm 3^\circ\text{C}$ determined through error analysis. Figure 10 illustrates a significant temperature increase in the laser beam impact zone A before nozzle passage, reflecting effective local preheating to ensure efficient molecular chain rearrangement at the interface between the previously deposited and melted filaments.

The local preheating temperature depends on (i) the laser intensity, (ii) the distance separating the optical system exit from the beam impact zone A, and (iii) the optical system speed relative to the solid block, which is equal to the printing speed. For the purpose of this study and considering the proposed experimental

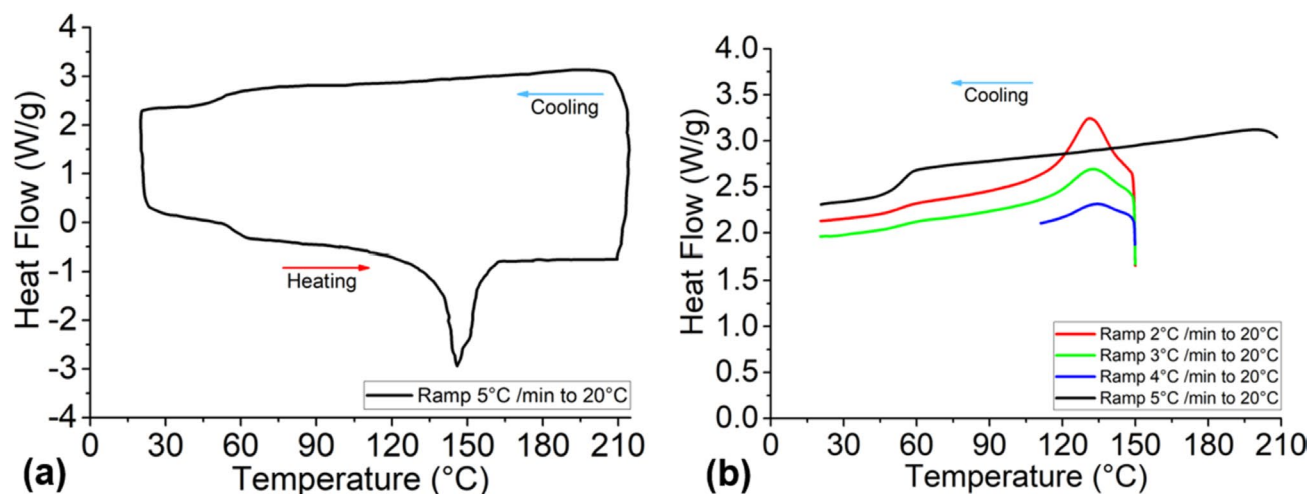


FIGURE 9 | DSC diagram of (a) heating and cooling of PLA with a rate of $5^{\circ}\text{C min}^{-1}$, and (b) the impact of cooling rate on crystallization rate.

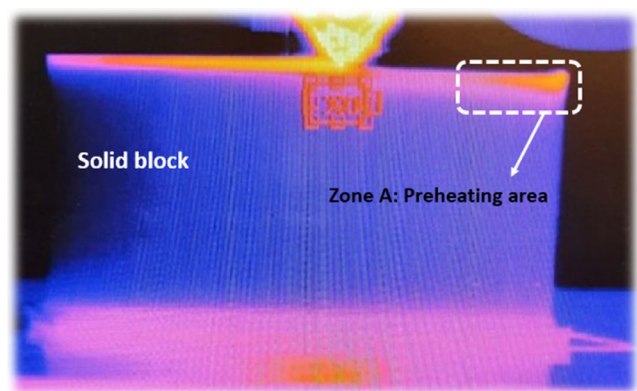


FIGURE 10 | 3D printing using a laser-based local preheating technique.

setup, the optical system speed was set to 20 and 40 mm min^{-1} and the distance between the optical system exit and the laser beam impact zone A was set to 80 mm , which corresponds to a laser spot diameter of approximately 0.8 mm . The experimental setup, as presented above, was used to print several solid blocks for different laser intensities ranging from 15 A to 35 A , and preheating temperatures were measured using a thermal camera and reported in Figure 11. It is observed from these results that as the laser intensity increases up to 35 A , the preheating temperature increases linearly to reach 270°C . It was also observed that for preheating temperatures higher than approximately 180°C , PLA solid block distortions were significant. Now that the relationship between laser intensity and preheating temperature is clarified, let us study the relationship between the local preheating temperature and the bonding strength.

5.3 | Laser Preheating Technique and Bonding Strength Improvement

The application of local preheating in solid block 3D printing aims to improve mechanical properties such as bond strength. To evaluate this improvement, solid blocks were printed using laser intensities generating local preheating temperatures ranging

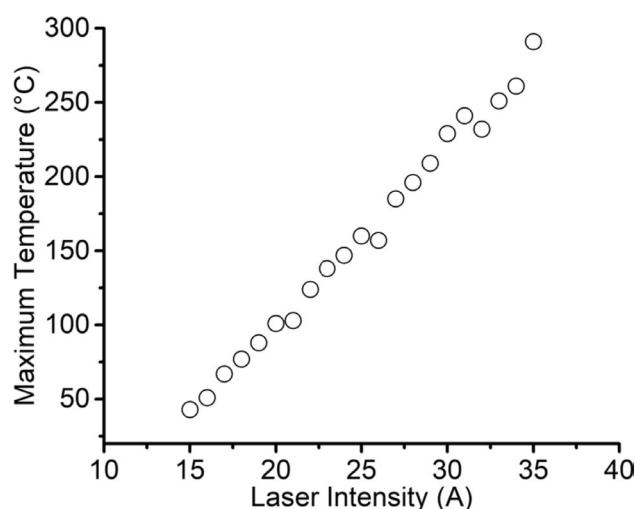


FIGURE 11 | Temperature variation versus laser intensity (80 mm). Note: Error bars represent measurement uncertainty of $\pm 3^{\circ}\text{C}$.

from 50°C to 180°C , followed by tensile tests. For each condition, three samples were tested to ensure statistical reliability. The results, presented in Figure 12 with mean bond strength values and standard deviations, demonstrate that the bond strength increases with local preheating temperature, which depends on laser intensity. Given the high cooling rate in FFF, where PLA remains largely amorphous (as explained in Section 5.1.2), these results could be attributed to improved adhesion between filaments due to enhanced molecular diffusion at higher temperatures. At 180°C , the mean bond strength increased by approximately 15% at a printing speed of 40 mm s^{-1} (from $20.0 \pm 3\text{ N}$ at 50°C to $23.0 \pm 3\text{ N}$) and by 35% at 20 mm s^{-1} (from $20.0 \pm 3\text{ N}$ to $27.0 \pm 3\text{ N}$), relative to the 50°C baseline. Structural integrity was maintained between 50°C and 180°C ; beyond 180°C , defects like distortions and filament fusion were observed.

Several factors beyond laser intensity influence these outcomes. Printing speed, tested at 20 and 40 mm s^{-1} , significantly affects results: slower speeds (20 mm s^{-1}) yield higher mean bond strength ($27.0 \pm 3\text{ N}$ vs. $23.0 \pm 3\text{ N}$ at 40 mm s^{-1} at 180°C), as shown in Figure 12, due to longer laser exposure

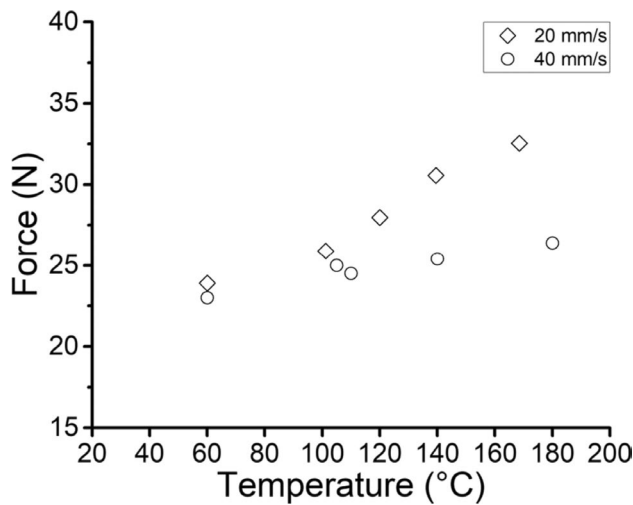


FIGURE 12 | Bonding strength versus local preheating temperature. *Note:* Standard deviations are estimated about ± 3 N.

(0.05 s vs. 0.025 s), allowing greater coalescence and macromolecular interdiffusion at the interface. PLA color also plays a role; darker colors (e.g., black, 75% absorptivity) absorb more laser energy than lighter ones (e.g., white, 10%–30%), potentially amplifying preheating effects and bond strength, though this study used black PLA for consistency (as noted in Section 4.1). Layer height (fixed at 0.2 mm) impacts heat transfer and contact area between filaments; thinner layers could enhance bonding by increasing surface interaction but may reduce throughput, while thicker layers might weaken adhesion due to reduced overlap, potentially increasing variability in strength measurements. Ambient conditions, such as platform temperature (set at 50°C) and humidity (30% during storage), further influence outcomes—higher ambient temperatures could reduce cooling rates, slightly aiding crystallization and consistency, while uncontrolled humidity might degrade PLA, though storage mitigated this risk. These factors interact with the preheating technique: higher absorptivity or slower speeds amplify temperature effects, while layer height and ambient conditions modulate thermal gradients and material behavior, as reflected in the standard deviations. Future studies could quantify these interactions to optimize outcomes further.

6 | Conclusions

This research investigates the contribution of the local preheating technique to FFF 3D printing. Combining the FFF 3D printing process with the laser local preheating technique allows for an improved bonding strength of printed parts. Using the designed laser-based setup, several PLA blocks were printed using different laser intensities and printing speeds. The tensile tests were conducted on the printed specimens. The obtained results show a significant increase in the bonding strength, up to 35%, without compromising the structural integrity. In addition, these results indicated that the bonding strength is proportional to the laser intensity/preheating temperature and inversely proportional to the printing speed.

This study has shown that the local preheating technique is a potential solution for improving the bonding strength of

3D printed parts. In future studies, it would be important to consider other sources of energy for local preheating, easier to handle than the laser, and design an annular preheating system, coaxial to the nozzle, allowing the printing of more complex parts.

Author Contributions

K.R. and H.R.V.: conception. **K.R. and F.C.:** experimental design. **A.G. and H.R.V.:** measurements. **A.T. and S.K.:** supervision and resources, contribution in manuscript writing: equal for all authors.

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Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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